

CAPSTONE PROJECT SUBMISSION

ACADEMIC INTEGRITY DECLARATION

AI/ML Fundamentals Course

This declaration must be completed and signed by the student. The project submission will not be considered complete without this confirmation.

Student Declaration

By signing this cover sheet, I confirm that:

- The submitted Capstone Project is my own work.
- I have not copied another student's project, code, report, documentation, or results.
- All external datasets, libraries, pretrained models, code samples, AI tools, and other sources used in the project are clearly acknowledged in the repository.
- I have not fabricated, altered, or misrepresented evaluation metrics, experiment results, screenshots, logs, or repository history.
- I have not presented an external AI service or hosted API as a model trained or fine-tuned by me.
- I understand the submitted code and can explain and defend the project, its pipeline, and its results.
- The repository link included in the submission form points to the final version of the project, contains the reproducible demo/Colab workflow, and is accessible to the mentors.

I understand that plagiarism, fabricated results, unauthorized collaboration, or misrepresentation of authorship may result in a score of 0 for the Capstone Project and may lead to additional academic disciplinary action.

[X] I have read, understood, and agree to the declaration above.

Full Name *	Feruzbek Baqoyev
Date *	16-08-2026
Student Signature *	I confirm

Complete this cover sheet before filling in the project submission form on the next page.

CAPSTONE PROJECT SUBMISSION FORM

AI/ML Fundamentals Course

Instructions: Download this file, complete all required fields, save it as PDF or DOCX, and upload it to the LMS. The repository must be accessible to the mentors before the deadline.

1. Full Name *	Feruzbek Baqoyev
2. Project Track *	☐ Individual Project Track ☒ Field-Based Scenario Track (FIN-01)
3. Project Title *	FIN-001: Automatic Transaction Categorization
4. Repository URL *	https://github.com/feruz-bb/automatic-transaction-categorization.git
5. Short Project Description *	Problem addressed: Raw bank transaction descriptions (e.g., "[debit] PUBLIX 7860", "MTG PMT PENFED CU") are noisy, abbreviated, and inconsistent. Manual categorization is expensive and unscalable, while rigid rule-based systems break when encountering new merchant formats. Expected user or stakeholder: Digital banking application users seeking automated personal spending breakdowns, and FinTech analytics/product teams needing reliable automated transaction categorization. ML task and main output: NLP multi-class text classification (TF-IDF + Logistic Regression across 17 spending categories). The system predicts the spending category, produces calibrated confidence probabilities with top-3 alternative rankings, and triggers a "needs_review" safety flag for ambiguous or zero-vocabulary unseen merchant transactions.
6. Optional Demo / Application Link	Not available (Demonstrated directly from the repository via local Streamlit app "streamlit run app.py" and automated verification test suite)
7. Access Instructions	Not required (Public GitHub repository with all dependencies documented in requirements.txt and pre-trained model bundle committed in artifacts/transaction_categorizer.joblib)

8. Repository Access Confirmation *

☒ I confirm that the repository is accessible to the mentors, contains the final project version, and includes the reproducible demo/Colab workflow or clear instructions to run it.